Week 11: Tools, agents and MCP

AI Development Course

Let the model use your code instead of guessing

Tested with anthropic 1.13.0 and claude-haiku-5-5 (real calls), mcp 2.3.0 with Python 3.12, and Claude Code 2.1.296's claude mcp commands (not signed in), 13 October 2026. A signed-in Claude Code session, GitHub, and Mac and Windows laptops are being tested.

Today

- Tool use: the model asks, your code runs
- An agent: that, in a loop, with a limit
- The deal screener's calculator as a tool
- Checking every input the model sends
- One safe tool in your project
- MCP: your tool, inside Claude Code
- Review, Docker, pull request

The tool cycle

1. Your program sends the question **and** the tool's description
2. The model replies: "please run `percent_change` with old 1500, new 1700"
3. **Your program** runs it and sends back `13.33`
4. The model answers with your number

The model never runs anything itself.

What goes back and forth

Claude's request (`stop_reason: "tool_use"`):

```
{"type": "tool_use", "id": "toolu_...", "name": "percent_change",
 "input": {"old": 1500, "new": 1700}}
```

Your reply (first in the next user message):

```
{"type": "tool_result", "tool_use_id": "toolu_...", "content": "13.33"}
```

Two calls, both paid: about **$0.00016** in our run.

An agent: the cycle in a loop

1. Send the question, the tool, the conversation so far
2. `end_turn`? Print the answer. Done.
3. `tool_use`? **Check the inputs**, run the tool, add the result, go to 1
4. Gone round 5 times? Stop and say so

The limit of 5 is the course's rule: Anthropic's docs don't set one.

Let code do the maths, let the model talk

The deal screener's one tool: `screen_listing`, the week 5 calculator.

```
Use the screen_listing tool for every number you report;
never work numbers out yourself.
```

Our run: `Monthly cash flow: $430.53`, exactly as in week 5. 2 turns, about **$0.00080**.

What the live runs taught

- First prompt: Claude said the calculator **left out** vacancy and repairs. (It doesn't.)
- Second: it rounded $430.53 to "about $431" and did its own sums.
- Fix: the tool's description names what's included; the prompt says "give the numbers exactly".

Test your prompt with real runs, and read the answers.

Never trust the inputs

The model can send anything. `screen_listing` checks every field:

Sent	Result
a missing field, words, a negative	refused, with the field named
a 250% rate, 30.5 years	refused
an interest rate of `0.07`	refused: "write 7 for 7%"

Refusals go back as `is_error: true`, so the model can fix it or ask you.

Missing numbers? Ask, don't guess

```
If the question is missing any of the nine numbers,
ask the user for them and do not call the tool.
```

Our run: it listed the seven missing numbers and called no tool (about **$0.00031**).

Safe first tools

Good this week	Not this week
A calculator you already have	Delete, send, pay, post, write files
Looking up your own course-safe data	Other people's accounts or data
Fixed conversions	Running code the model writes

Week 13 is planned to cover what can go wrong with more powerful tools.

MCP in one picture, in words

Claude Code speaks **MCP** to **your MCP server**, which calls **your function**.

- MCP: a standard way to package tools for AI apps
- Write the tool once; Claude Code can use it while it works

The server

```
from mcp.server import MCPServer

mcp = MCPServer("dealscreener")

@mcp.tool()
def screen_listing(...):
    ...same checks, same calculator...
```

- `mcp[cli]==2.3.0`, Python **3.11 or newer**
- Old tutorials say `FastMCP`: that name is gone in 2.3.0

Connect it to Claude Code

```
claude mcp add dealscreener --scope project -e PYTHONPATH="$PWD" -- "$PWD/.venv/bin/python" -m dealscreener.mcp_server
claude mcp list
```

- In a **new window with no key loaded**: Claude Code's servers would get it
- `--scope project`: saved in `.mcp.json`, this project only
- `-e PYTHONPATH="$PWD"`: Python can find your code from anywhere
- Full path to the `.venv`'s Python: saved exactly as typed
- "Pending approval": approve it when you start `claude` (being tested)

Trust before you connect

Claude Code's security page:

> "We encourage either writing your own MCP servers or using MCP servers from providers that you trust."

Anthropic "does not security-audit or manage any MCP server".

This week: connect only the server **you** wrote.

Your tool, step by step

1. Branch `tool` first
2. Pick a read-or-calculate tool
3. `/superpowers:writing-plans` with the guardrails, then build
4. Ask three questions: answerable, missing input, bad input
5. Tests pass with `.env` moved out
6. MCP server, `claude mcp add`, a screenshot of Claude Code calling it

Tests that never call Anthropic

- A fake model that **asks for the tool**
- A bad input, and two tool calls in one reply
- Hitting the 5-turn limit
- One live test, only with `RUN_LIVE_TESTS=1`
- The deal screener: 184 tests

The assignment

Criterion	Points
Merged PR: read-or-calculate tool, capped loop, checked inputs, fake-model tests, no key	4
MCP server connected: command, screenshot, one agent run's output	3
Reflection: when to call a tool, and what could go wrong with a tool that changes things	3

Next week: evaluation.